Algebraic Curves

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There should be three sections: Projective geometry, Intersections of curves, Riemann surfaces

Contents

1 Affine varieites

Throughout, let k be a field.

Definition 1.0.1. For a finite collection of polynomials $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ we define the **affine algebraic variety**[1] $\mathbb{V}_k(f_1, \dots, f_m)$ as the set of common zeroes of all the polynomials in k^n.

[1] There are complexities to this deifnition like irreducibility, but this is an introductory course so I won't focus on that notation.

So many basic objects of geometry are captured as affine algebraic varieties: finite sets of points $\mathbb{V}_k(x_1-a_1,\ldots,x_n-a_n)$, conics $\mathbb{V}_{\mathbb{R}}(ax^2+bxy+cy^2+dx+ey+f)$, affine spaces $\mathbb{A}^n_k := \mathbb{V}_K(0)$, and many others. We will restrict our scope of study in this course.

Definition 1.0.2. An **affine plane curve** over k is the affine variety:

$$C = \mathbb{V}_k(P) := \{(x,y) \in k^2 \mid P(x,y) = 0\}$$

where P is a non-constant polynomial in two variables.

There are some natural first questions we can ask about affine plane curves:

- How does this geometrically represent the factorisation of our plane curve?
- How can we compute the number of intersections of two plane curves?
- When to two polynomials give the same plane curve?
- Can we classify all plane curves over certain fields?
- What is the appropriate notion of two curves being equivalent?

We'll begin to develop some of the basic theory of algebraic geometry so that we can state these questions more precisely.

Theorem 1.0.3 (Hilbert basis theorem)**.** If R is a Noetherian ring, then so is $R[X]$

Proof. This is in my notes for algebra 3. $\square$

Corollary 1.0.4. All ideals $I \trianglelefteq k[x_1,\ldots,x_n]$ are finitely generated.

So I can expand the definition of an affine variety.

Definition 1.0.5. The **affine variety** associated to an ideal $I \trianglelefteq k[x_1,\ldots,x_n]$ is:

$$\mathbb{V}(I) := \{(a_1,\ldots,a_n) \in k^n \mid f(a_1,\ldots,a_n) = 0 \text{ for all } f \in I\}$$

The Hilbert basis theorem tells us any such I will be finitely generated, i.e.

$$I = (f_1, f_2, \ldots, f_m)$$

so we should have

Lemma 1.0.6. $\mathbb{V}(I) = \mathbb{V}(f_1,\ldots,f_m)$.

Proof. ($\subset$) Take $(a_1,\ldots,a_n) \in \mathbb{V}(I)$, certianly each of the $f_i \in I$, so $(a_1,\ldots,a_n)$ vanishes an all f_i and is thus in $\mathbb{V}(f_1,\ldots,f_m)$.

($\supset$) Now suppose $(a_1,\ldots,a_n) \in \mathbb{V}(f_1,\ldots,f_m)$, we know for any $f \in I$, there exists $g_1,\ldots,g_m$ such that

$$f = g_1f_1 + g_2f_2 + \cdots g_mf_m$$

as each f_i is zero at $(a_1,\ldots,a_n)$ so is f. $\square$

Lemma 1.0.7. The set of affine varieties is closed under finite unions and arbitrary intersections.

Proof. Let $V_1 = \mathbb{V}(f_1,\ldots,f_m)$ and $V_2 = \mathbb{V}(g_1,\ldots,g_l)$, then I claim

$$V_1 \cup V_2 = \mathbb{V}(f_ig_j \mid \text{for all } 1 \leq i \leq m,\ 1 \leq j \leq l)$$

($\subset$) Certainly if $(a_1,\ldots,a_n) \in V_1 \cup V_2$ then it will be zero for all f_i or all g_j, so will always be zero on all products f_ig_j.

($\supset$) If, instead, $a = (a_1,\ldots,a_n)$ is zero for all f_ig_j and, wlog, a is nonzero for some f_i, then as $f_ig_1, g_ig_2, \ldots, f_ig_l$ all have a as a zero, and $k[x_1,\ldots,x_n]$ is an integral domain, $a \in V_2$.

Now suppose $\{\mathbb{V}(I_\alpha)\}_{\alpha\in A}$ is a set of varieties indexed by some set A, let $I = \bigcup_{\alpha\in A} I_\alpha$, I claim:

$$\bigcap_{\alpha\in A} \mathbb{V}(I_\alpha) = \mathbb{V}(I)$$

Certianly if a is in all $\mathbb{V}(I_\alpha)$, then it is a zero of all polynomials in each I_α, which is precisely the polynomials in I. If $a \in \mathbb{V}(I)$, then by the Hilbert basis theorem I is finitely generated by some polynomials $f_1,\ldots,f_m$ appearing in $I_{\alpha_1},\ldots I_{\alpha_m}$ respectively, we know these polynomials generate I, so in fact generate all ofe very I_α, so a is in each $\mathbb{V}(I_\alpha)$. TODO: terrible proof. $\square$

Theorem 1.0.8 (Fundamental theorem of algebra). $\mathbb{C}$ is algebraically closed.

Proposition 1.0.9. If k is algebraically closed, then any affine plane curve over k has infinitely many points.

Proof. □

1.1 Nullstellensatz

Theorem 1.1.1 (Weak Nullstellensatz). For k an uncountable algebraically closed field, and $I \trianglelefteq k[x_1, \dots, x_n]$, the variety $V(I)$ is empty iff $1 \in I$.

Proof. If $1 \in I$, then $V(I)$ will definitely be empty, so we suppose $1 \neq I$ and show $V(I)$ is nonemtpy.

By Zorn's lemma, we know there exists some maximal ideal $\mathfrak{m}$ such that $I \subseteq \mathfrak{m}$, so it suffices to show that the variety of a maximal ideal is nonempty. We prove the slightly stronger claim, that any such $\mathfrak{m}$ is of the form:

$$\mathfrak{m} = (x_1 - a_1, x_2 - a_2, \dots, x_n - a_n) \trianglelefteq k[x_1, \dots, x_n]$$

for some $a = (a_1, \dots, a_n) \in k^n$. In this case, $V(\mathfrak{m})$ is exactly a.

Since $\mathfrak{m}$ is maximal, $k[x_1, \dots, x_n]/\mathfrak{m}$ will be a field extension of k, if this is isomorphic to k, then the image of the variables in the projection are the $a_1, \dots, a_n$ in our desired form and we are done.

If, instead, $K = k[x_1, \dots, x_n]$ is a bigger field than k, it will be algebraic and thus countable generated as a k-vector space. For some $t \in K \setminus k$ consider the uncountable set of $1/(t - a)$ for all $a \in k$, this will have to be linearly dependent, so we have a finite sum:

$$\sum_{i=1}^{n} \frac{b_i}{t - a_i} = 0 \implies f(t) := \sum_{i=1}^{n} b_i \prod_{j \neq i} (t - a_j) = 0$$

For distinct a_i and possible non-distinct b_i all in k. This expresses t as the root of a polynomial $f \in k[x]$, we know $f(a_1)$ is nonzero as the a_is are distsinct, so f is not the zero polynomial. But as k is algebraically closed, we have reached a contradiction. □

Theorem 1.1.2 (Strong Nullstellensatz). For k and I as above, the ideal of polynomials which all vanish on all points of $V(I)$ is exactly $\sqrt{I}$.

Proof. By the Hilbert basis theorem we write $I = (f_1, \dots, f_m)$, if we have $g \in k[x_1, \dots, x_n]$ vanishing whenever all the $f_1, \dots, f_m$ do, we have to show there exists some $r > 0$ such that $k^r \in I$. First observe that in the ring $k[t, x_1, \dots, x_m]$, the set of polynomials $f_1, \dots, f_m, 1 - tg$ have no common zeroes, and so by the Weak Nullstellensatz they generate the unit ideal, thus we can find $h, h_1, \dots, h_{m+1} \in k[t, x_1, \dots, x_m]$ such that:

$$h(1 - tg) + \sum_{i=1}^{m} h_i f_i = 1$$

Let r be the highest power of t that appears in any of the h_is and consider the projection of:

$$g^r h(1 - tg) + g^r \sum_{i=1}^{m} h_i f_i = g^r$$

into the ring $k[t, x_1, \dots, x_n]/(tg - 1)$. This expresses g^r as a $k[x_1, \dots, x_n]$-linear combination of the f_is, modulo the ideal generated by some polynomial with nonzero t term. Thus they are equal in $k[x_1, \dots, x_n]$. □

The method of bringing in the new variable t in this proof is called the "Rabinowitsch trick".

Corollary 1.1.3. For k as above, $f, g \in k[x_1, \dots, x_n]$, f is irreducible, and $V(f) \subseteq V(g)$, then f divides g.

Proof. By the strong Nullstellensatz at $I = (f)$, we know there exists some r such that $g^r = af$ for some $a \in k[x_1, \dots, x_n]$, as f is irreducible and $k[x_1, \dots, x_n]$ has unique factorisation, f must divide g. □

9 Riemann-Roch

This section is the optional mastery material, the material is from Kirwan and Fulton.

Definition 9.0.1. For f, g meromorphic functions on a Riemann surface S, we call

$$f\ dg$$

a meromorphic differential on S. If $\tilde{f}$ and $\tilde{g}$ are also meromorphic functions on S we write $f\ dg = \tilde{f}\ d\tilde{g}$ iff for ever holomorphic chart $\phi : U \to V$ of S, we have

$$(f \circ \phi^{-1})(g \circ \phi^{-1})' = (\tilde{f} \circ \phi^{-1})(\tilde{g} \circ \phi^{-1})'$$

9.1 Divisors

Throuhgout, let C be a nonsingular projective plance curve with genus g.

Definition 9.1.1. The group of **divisors** on C is $\mathrm{Div}(C)$ is the free abelian group on the points of C. For $D, D' \in \mathrm{Div}(C)$, we use the ordering $D \geq D'$ if every term of $D - D'$ is ≥ 0. Such a divisor is called **effective**. The **degree** of a divisor is the sum of its coefficients in $\mathbb{Z}$.